

1.7 Writing and Solving Equations in Real-World Context

Lesson Introduction

 Benchmark	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.		 Learning Targets	- I can write two-step equations to represent real-world contexts. - I can solve two-step equations within a real-world context.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.		MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.	
 Essential Questions	- How do you write a two-step equation to represent a real-world context? - How do you solve a two-step equation for a real-world context?			
 Lesson Narrative	In this lesson students are guided through writing and solving two-step equations in the form $px + q = r$ and $p(x + q) = r$. Students apply their knowledge and skills from the previous lesson to accurately and efficiently write and solve two-step equations in real-world contexts.			
 Component Summary	1.7.1 Warm-Up (~5 min.)	Notice and Wonder about a scatter plot graph (<i>SE p. 42</i>)	1.7.3 Bring It Together (5 min.)	Reflect on learning over (<i>SE p. 45</i>)
	1.7.2 Collaboration (25 min.)	Collaboratively write and solve equations in real-world contexts (<i>SE p. 42</i>)	1.7.4 Your Turn! (5 min.)	Independent practice writing and solving two-step equations in real world contexts (<i>SE p. 45</i>)
	1.7.5 Wrap-Up (~5 min.)	Self-assessment of learning targets (<i>SE p. 46</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Calculators (Optional) Chart paper and markers for gallery walk	
			Additional Preparation	
			Consider creating and displaying an anchor chart with the steps needed to write and solve two-step equations from real-world problems.	

 Teacher Tips	Common Misconceptions • In 1.7.2 Collaboration problem 3, students may struggle to determine the inverse operation of multiplication by $\frac{1}{8}$. Students may attempt to divide by 8 to solve. • Students may think that all equations will be in the form of $px + q = r$ and just insert values in for p, q, and r.	Things to Consider • If time is an issue, then consider assigning Your Turn! for homework. • Consider having no-calculator questions where students turn their calculator over. Consider allowing students to use calculators on the problems where the burden of calculations could distract students from focusing on reading and interpreting the real-world contexts. • Consider using observations and student work to determine which students could benefit from small-group intervention. Then, utilize small-group intervention during the Collaboration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 1.7.2 Collaboration, students collaboratively define the variable, write an equation, solve the equation, and interpret the solution of a real-world problem. Ask students to take turns completing each row in the table while their partner agrees or disagrees, and/or asks clarifying questions as they complete the problems.	MA.K12.MTR.4.1: Discussion and Reflection In 1.7.2 Collaboration, students collaboratively write and solve five different equations. They discuss their work and the meaning of the solutions with a partner. This activity provides students with opportunities to reflect on their own and the mathematical thinking of others.
 Differentiate Instruction	Consider putting students in groups and assigning groups different problems in 1.7.2 Collaboration. Then, have students share their work on a poster during a gallery walk. Structuring this activity as a jigsaw will provide students with auditory and visual entry-points.	
Lesson Notes:		

1.7.1 Warm-Up: Bell Work

Component Overview: Students write and solve an equation that models attendance at Bethune-Cookman University. Consider engaging students into a discussion about this Historically Black College and/or Florida's four Historically Black Colleges.



1.7.1 Warm-Up: Bell Work

Mary McLeod Bethune was an educator, civil rights activist, and advisor to five U.S. presidents. She turned her passion for the U.S. making racial progress into founding the Daytona Literacy and Industrial Training School for Negro Girls in 1904. Twenty years later, the school would merge with Cookman Institute of Jacksonville to create what is known today as Bethune-Cookman University, a Historically Black College University (HBCU). Mary McLeod Bethune was president of the university until 1942.



1. In 2021, there were 3,773 students enrolled at Bethune-Cookman University, which is 3,648 more than 25 times the number of students enrolled in 1904.

- a. Write an equation that could be used to find the number of students enrolled in 1904, n .

$$3773 = 25n + 3648$$

- b. Solve your equation.

$$3773 - 3648 = 25n + 3648 - 3648$$

$$125 \div 25 = 25n \div 25$$

$$n = 5$$



Consider inviting an alumnus from Bethune-Cookman University or another Historically Black College to speak to your class.



There are four Historically Black Colleges in Florida.

- Bethune-Cookman University in Daytona Beach
- Edward Waters College in Jacksonville
- Florida A&M University in Tallahassee
- Florida Memorial University in Miami Gardens

Consider asking students what they know about these colleges.

1.7.2 Collaboration: Writing and Solving Equations in Real-World Contexts

Component Overview: Students collaboratively write and solve two-step equations in real-world contexts. There are different methods of solving equations. Provide students with the flexibility to write and solve equations by selecting a procedure that works best for them.



1.7.2 Collaboration: Writing and Solving Equations in Real-World Contexts

Work with your partner or small group to complete the table for each real-world context.

- Peyton was given a \$40 gift card to her local coffee shop. She went to the coffee shop 4 times and purchased the same thing each time. If she has \$7.08 left on her gift card, what was the cost of her purchase each time she went to the coffee shop?

Define the variable.	$c = \text{the cost of each purchase made}$
Write an equation that can be used to answer the question.	$40 - 4c = 7.08$
Solve the equation.	$ \begin{array}{r} 40 - 4c = 7.08 \\ 40 - 40 - 4c = 7.08 - 40 \\ -4c = -32.92 \\ \frac{-4c}{-4} = \frac{-32.92}{-4} \\ c = 8.23 \end{array} $
Explain the meaning of the solution.	$\text{Peyton spent \$8.23 each time she went to the coffee shop.}$

- Florida Power and Light (FPL) charges customers 22.358 cents per kilowatt-hour of electricity used each month. Monthly charges to customers also include \$18.75 per month for service. If a customer receives a bill for \$237.86, how many kilowatt-hours were used during the month? Round your answer to nearest tenth.

Define the variable.	$k = \text{the number of kilowatt-hours used this month}$
Write an equation that can be used to answer the question.	$0.22358k + 18.75 = 237.86$



There are different methods to writing and solving equations. Provide students with the flexibility to write and solve equations by selecting a procedure that allows them to both write and solve efficiently.



Consider putting students in groups and assigning groups different problems. Then, have students share their work on a poster during a gallery walk. Structuring this activity as a jigsaw will provide students with auditory and visual entry points.



Students may struggle with the context because they do not have experience with electric bills. Having students solve problems where the context is not familiar to them often requires students to use a growth mindset. Encourage students who would prefer to give up because they do not understand the context.

1.7.2 Collaboration: Writing and Solving Equations in Real-World Contexts (continued)

Solve the equation.	$0.22358k + 18.75 = 237.86$ $0.22358k + 18.75 - 18.75 =$ $237.86 - 18.75$ $0.22358k = 219.11$ $\frac{0.22358k}{0.22358} = \frac{219.11}{0.22358}$ $k = 980.0$
Explain the meaning of the solution.	<i>980 kilowatt-hours were used this month by this customer.</i>

3. Kai is making ham and cheese sandwiches for an upcoming camping trip with his family. He uses $\frac{1}{8}$ pound of cheese for each sandwich. He has a 2-pound block of cheese and needs to use 0.75 pounds of it for a different recipe. How many ham and cheese sandwiches will he be able to make?

Define the variable.	<i>s = the number of sandwiches Kai can make</i>
Write an equation that can be used to answer the question.	$\frac{1}{8}s + 0.75 = 2$
Solve the equation.	$\frac{1}{8}s + 0.75 = 2$ $\frac{1}{8}s + 0.75 - 0.75 = 2 - 0.75$ $\frac{1}{8}s = 1.25$ $8 * \left(\frac{1}{8}s\right) = 8 * (1.25)$ $s = 10$
Explain the meaning of the solution.	<i>Kai will be able to make 10 sandwiches.</i>



Students may struggle with how to perform the inverse operation of multiplying by $\frac{1}{8}$. They may attempt to divide by 8 to solve. Show students how multiplying by a unit fraction is equivalent to dividing by the denominator. Then, ask students what the inverse operation of division is.

1.7.2 Collaboration: Writing and Solving Equations in Real-World Contexts (continued)

4. Fatima and Luna are members of the running club at school. Fatima ran 7 fewer miles this week than she did last week. She ran 9 times as far as Luna ran this week. Luna only had time to run 4 miles this week. How many miles did Fatima run last week?

Define the variable.	$m = \text{the number of miles Fatima ran last week.}$
Write an equation that can be used to answer the question.	$m - 7 = 9 * 4$
Solve the equation.	$m - 7 = 9 * 4$ $m - 7 = 36$ $m - 7 + 7 = 36 + 7$ $m = 43$
Explain the meaning of the solution.	$\text{Fatima ran 43 miles last week.}$

5. Shirley gives each of her 8 grandchildren a \$15 gift card to their favorite restaurant and some cash for their birthday. She spends the same amount of money on each grandchild. If she spends \$280 each year on her grandchildren's birthday gifts, how much cash does she give each grandchild on their birthday?

Define the variable.	$c = \text{the amount of cash Shirley gives each of her}$
Write an equation that can be used to answer the question.	$8(15 + c) = 280$
Solve the equation.	$\frac{8(15 + c) = 280}{8(15 + c) \quad 280}$ $\frac{8}{8} = \frac{8}{8}$ $15 + c = 35$ $15 - 15 + c = 35 - 15$ $c = 20$
Explain the meaning of the solution.	$\text{Shirley gave each of her grandchildren \$20.}$



Students may struggle with where to put -7 in the equation and may write $7 - m$ when they see 7 fewer miles. Those students will have a difficult time explaining the meaning of the solution. Allow these students to converse with others who wrote $m - 7$ and discuss why they wrote it that way. Then ask students why $m - 7$ makes sense for the context of the problem.



Remind students to check their work by considering whether or not the answer makes sense in the context and by using substitution.

1.7.3 Bring It Together: Writing and Solving Equations in Real-World Contexts

Component Overview: Students reflect on errors they have made when writing equations to represent real-world contexts. Then, students share their errors with a partner. The goal of this activity is to help students avoid making similar errors in the future and to use a growth mindset.



1.7.3 Bring It Together: Writing and Solving Equations in Real-World Contexts

Consider some of the problems you've solved and errors you've made over the last few lessons.

1. In the first column of the table, write yourself a note or reminder that you can use when writing an equation for a real-world context so you can avoid similar errors in the future.

My Note or Reminders	My Partner's Reminders
<i>Answers will vary.</i>	<i>Answers will vary.</i>

2. Share your list with your partner. Add any helpful reminders your partner shares in the second column.
Answers will vary.



Think-Pair-Share

Have students initially record their reflections independently. Then, have students share their reflections with their partners.

If time allows, consider showing students examples of student errors you have noticed. This may help students consider errors they have made.

1.7.4 Your Turn!

Component Overview: Students independently write and solve a two-step equation in a real-world context. If students struggle to write the equation, ask them to underline important quantities in the problem. Remind students to check their work after completing the problem.



1.7.4 Your Turn!

- Complete the table for the real-world situation.

Kiara is baking cookies for an upcoming bake sale. Each batch of cookies requires $1\frac{2}{3}$ cups of sugar. She has 12 cups of sugar, but her mom asked her to save $\frac{1}{3}$ cup for her to use in her coffee until she can go to the grocery store in a few days. How many batches of cookies can Kiara make

Define the variable.	$b = \text{batches of cookies}$
Write an equation that can be used to answer the question.	$1\frac{2}{3}b + \frac{1}{3} = 12$
Solve the equation.	$1\frac{2}{3}b + \frac{1}{3} = 12$ $1\frac{2}{3}b + \frac{1}{3} - \frac{1}{3} = 12 - \frac{1}{3}$ $1\frac{2}{3}b = 11\frac{2}{3}$ $\frac{1\frac{2}{3}b}{1\frac{2}{3}} = \frac{11\frac{2}{3}}{1\frac{2}{3}}$ $b = 7$
Explain the meaning of the solution.	$Kiara \text{ can make } 7 \text{ batches of cookies.}$



If students finish early, ask them to write their own real-world word problem that can be represented with a two-step equation. Then, have students exchange problems with a partner. Partners can write and solve the equations that represent each other's real-world word problems.

1.7.5 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment by reflecting on their mastery. Utilize this formative assessment to determine which students need remediation before continuing this unit.



1.7.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each item.

Answers will vary.



I need help.



I got it.



I could help someone.

Solving two-step equations			
Writing two-step equations to represent real-world contexts			
Explaining the meaning of solutions to equations in context			



For additional information, prompt students to use these symbols to mark-up the activities from the lesson.